

MCQs:

1. B
2. D
3. C
4. C
5. C
6. C
7. C
8. B
9. B
10. C
11. B
12. B

Q: 2(i) (or)

A: 2(i) (or)

Liver cells:

Are almost round in shape, with a prominent nucleus without cytoplasmic organelles. This makes ~~an~~ them the most metabolically active cells in the body. Here are two functions:

- ① Detoxification of toxic substance
- ② Recycling of old red blood cells.

Q: 2(ii)

A: 2(ii)

All living organism shares same
(7) characteristics :-

- ① Movement
- ② Respiration
- ③ Sensitivity
- ④ Excretion
- ⑤ Reproduction
- ⑥ Nutrition
- ⑦ Growth

Q: 2(iii)

A: 2(iii)

- A. Nucleus
- B. Mitochondria
- C. Golgi Apparatus
- D. Ribosomes

Q: 2(iv) (or)

A: 2(iv) (or)

Functions :-

① Haemoglobin - transports oxygen and CO_2 in blood.

② Keratin: hardens hair, nails, feathers, horns, beaks.

③ Histone :- Attaches to DNA from chromosomes.

④ Antibodies :- provide immunity against germs.

Q: 2(v) (or)

A: 2(v) (or)

Most enzymes are made up of chains of amino acids to form 3D proteins are known as globular structure of enzymes. A few enzyme are RNA in nature is called ribozymes.

Q. 2(vi)(or)

A. 2(vi)(or)

Upper Epidermis:-

is a single layer of cells covered by a waxy substance called cutin, which forms the cuticle. This protects the leaf tissues and prevents loss of water.

Q. 2(vii)(or)

A. 2(vii)(or)

A monomer is a single, simple building-block molecule e.g. glucose.

A polymer is a large, complex molecule built from many repeating monomer units bonded together. E.g. starch.

Q: 2 (viii) (or)

A: 2 (vi) (or)

Qualitative
observations

Quantitative
Observations

① Subjective method

① Objective method

② Quality of things

② Quantify things

③ Not accurate
and not measure-
ble kind of
observations.

Accurate and measurable
Kind of observations

④ Use of five
senses.

Use of measuring
instruments.

Example:-
A leaf is green

Example:-
A length is '8 cm'

Q: 2 (in)

A: 2 (in)

① Plant sources :-

beans, lentils

② Animals sources :-

meats, milk.

Q: $2(n)$ (or)

A: $2(n)$ (or)

The mitotic phase is the part of cell cycle during which the cell actually divides to form daughter cells.

3 reasons :-

development

growth

sexual/asexual reproduction

Q: $2(n)$ (or)

A: $2(n)$ (or)

Systematics :-

is the scientific study of the diversity of organisms and their evolutionary relationships.

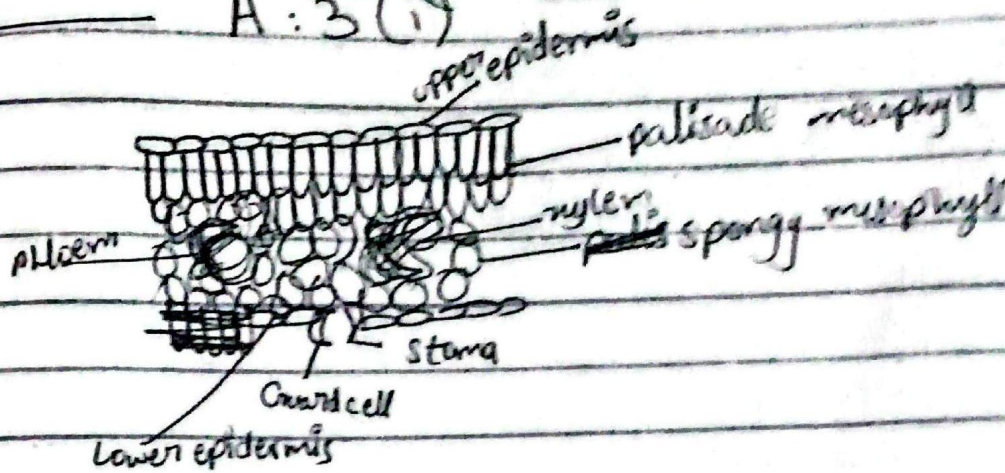
Taxonomy :-

Taxonomy is the branch of ~~branch~~ biology that is naming, identification and classification of organisms.

Long Question

Q: 3(i)

A: 3(i)



upper epidermis

lower epidermis

Phloem

Xylem

palisade mesophyll

spongy mesophyll

stoma

Guard cell .

Q: 3(ii)(or)

A: 3(ii)(or)

Seven characteristics of Life :-

① Movement

② Respiration

③ Sensitivity

④ Excretion

⑤ Nutrition

⑥ Growth

⑦ Reproduction .

Eight taxonomic hierarchy :-

Domain

Kingdom

Phylum / division

Class

Order

Family

Genus

Species.

Q. 3(iii)

A. 3(iii)

(a) Collagen / Keratin :-

Collagen is a major component in the structure of skin, bones, cartilage, muscles, tendons, ligaments. providing support and strength.

Keratin is the components of hair, nail, feather, horns and beaks. Making them hard and protective.

(b) Actin/myosin :-

These proteins are presents in muscle cells and are involved in the contraction and relaxation of muscles during movement. E.g locomotion, breathing movements, blood pumping by heart, peristaltic contraction of gut and speech movements.

(c) Haemoglobin :-

RBCs contains red pigment haemoglobin, a protein that transports oxygen and carbon dioxide in the body.

(d) Antibodies :-

Antibody proteins produced by white blood cells.

Q:3(iv)

A:3(iv)

(a) Four fat-soluble vitamins mentioned in the textbook

are :-

vitamin A, D, E, K - They are lipids in nature. Vitamin D

is synthesized/derived from
~~the~~ cholesterol in the body.

Steroids are lipids that do
not have fatty acids in their
structure. ~~Instead~~ instead, they
have distinctive 3-ring fused
ring structure.

Example :-

Cholesterol
Vitamin D.